MACHINE LEARNING (ML) / AI IN PUBLIC SECTOR / CRITICAL DATA STUDIES (CDS)

PREDICTING HIGH-RISK CRIME AREAS IN TARTU

A synthetic data proof-of-concept for ML-based decision support

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PROBLEM

Police resource allocation is often reactive, while real crime data is restricted by privacy, ethics, and access rules.

Objective: test whether a simple ML pipeline can predict high-risk area-time combinations without using real personal data.

Core question: can predictive policing systems be built technically, and what risks emerge when they guide decisions?

METHOD

Unit of analysis: one Tartu district at one hour. Output: probability of the area-hour being high-risk.

Synthetic data > Feature engineering > ML comparison > Model choosing > Application

DATA

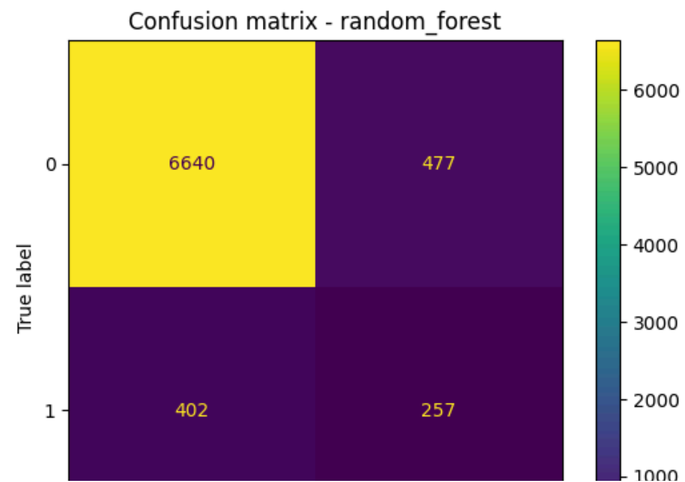
Environment	18 districts x 90 days x 24 hours = 38,880 area-hours.
Target	High-risk = incident_count >= 1; 3,297 rows (8.48%) are high-risk.
Time	hour, weekday, weekend, night

RESULTS

Random forest was the strongest high-risk classifier by F1. Accuracy is high because most area-hours are normal; precision shows how reliable a risk flag is, recall shows how many high-risk hours are caught, and F1 balances both.

88.7% Random forest accuracy	**0.369** Random forest F1	**0.812** Best ROC-AUC, logistic
0.350 Random forest precision	**0.390** Random forest recall	**38,880** Area-hour rows

CONFUSION MATRIX RANDOM FOREST



LIMITATIONS

- Synthetic data simplifies real-world complexity.
- The model predicts patterns, not causes.
- Inequality and segregation are missing or simplified.
- Thresholds turn uncertainty into concrete action.

APPLICATION

- Interactive Tartu risk map with district-level scores.
- Selected suburb view shows 48-hour predicted incidents and risk.
- Prototype links model details, dataset view, policing plan, and critical analysis.

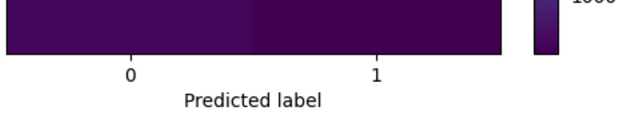
Context	weather and city events
Area	density, nightlife, student zones
History	past incident patterns

Why synthetic?	No privacy risk, full control, reproducible experimentation.
Weak point	The model learns the assumptions used to create its reality.

MODELS TESTED

Model	Task	Main setup	Notebook result
Linear regression	Count prediction	One-hot encoded features; 80/20 train-test split.	MAE 0.164, MSE 0.105, R2 0.146. Weak fit for incident counts.
PyCaret regression sweep	Count prediction	19 regressors, 10-fold CV, session 845; best was gradient boosting.	Best R2 0.197, RMSE 0.316, MAE 0.149. Counts remained hard to explain.
Decision tree	Incident-count class	Default DecisionTreeClassifier, no depth limit.	Accuracy 0.894, macro-F1 0.225. Mostly learned the zero-incident majority.
Logistic regression	High-risk binary	Balanced class weights, max_iter 1000, probability scores.	Acc 0.749, precision 0.217, recall 0.753, F1 0.337, ROC-AUC 0.812.
Random forest	High-risk binary	100 trees, min_samples_leaf 2, balanced_subsample, thresholded labels.	Acc 0.887, precision 0.350, recall 0.390, F1 0.369, ROC-AUC 0.768. Best F1.

Takeaway: classification was more useful than count regression; random forest gave the best F1, while logistic regression ranked risk best by ROC-AUC.



How to read: Rows show the real label; columns show the model prediction. Label 0 means normal, label 1 means high-risk. Correct predictions are 6,640 normal area-hours and 257 high-risk area-hours. Errors are 477 false alarms and 402 missed high-risk cases.

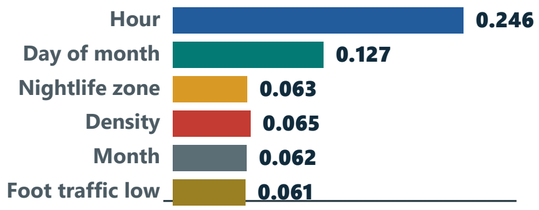
ROC CURVE

ACTUAL SCORES

ROC reading: The ROC curve shows the trade-off between catching high-risk area-hours and falsely flagging normal ones. Logistic regression ranked cases best overall (ROC-AUC 0.812); random forest was lower on ROC-AUC (0.768) but performed best on F1, so it is used for the confusion matrix above.

FEATURE IMPORTANCE

RANDOM FOREST



Feature insight: Top features are time and area descriptors: strong signals, strong assumptions.

CRITICAL ANALYSIS

Operational logic: identify top-risk district-hours and allocate more patrol attention there.

More patrols > **More detected incidents**

Higher risk score > **More patrols again**

Critical point: the system does not detect crime; it detects where police look.

- **Datafication:** crime becomes a pattern, not a social phenomenon.
- **Feedback loops:** repeated attention can manufacture hotspots.
- **Bias amplification:** data structures can reproduce inequality.
- **False objectivity:** scientific-looking outputs hide assumptions.

CONCLUSION

Best model: random forest was the most useful classifier for high-risk area-hours.

Main limitation: incident-count prediction was weak because high-risk rows are rare.

ML takeaway: engineered time, area, context, and history features can rank synthetic risk better than chance, but the prototype needs stronger validation before any real-world use.

References

Project materials in GitHub: Projekt (1).ipynb; tartu_risk_ml_models-1.ipynb; A1 poster

Critical frame: Meijer, A., & Wessels, M. (2019). Predictive Policing: Review of Benefits and Drawbacks. International Journal of Public Administration, 42(12), 1031-1039. https://doi.org/10.1080/01900692.2019.1575664